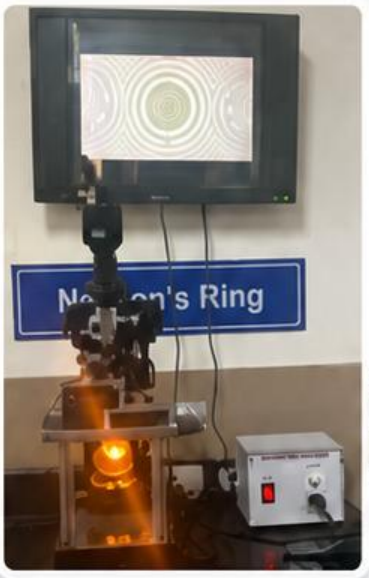
MEDICAPS UNIVERSITY

















APPLIED PHYSICS LAB MANUAL

Faculty of Science
Department of Physics

2.1 Objective: To simulate and analyze a charged particle's spiral trajectory and energy gain in a cyclotron.

2.2. Apparatus/Components Required: Personal Computer and Simulation software.

2.2. Theory:

A cyclotron is a machine that accelerates charged particles or ions to high energies. It was invente10.2d to investigate the nuclear structure by E.O Lawrence and M.S Livingston in 1934. Both electric and magnetic fields are used in the cyclotron to increase the energy of the charged particles. As both the fields are perpendicular to each other, they are called cross fields.

In a cyclotron, charged particles accelerate outwards from the centre along a spiral path. These particles are held to a spiral trajectory by a static magnetic field and accelerated by a rapidly varying electric field.

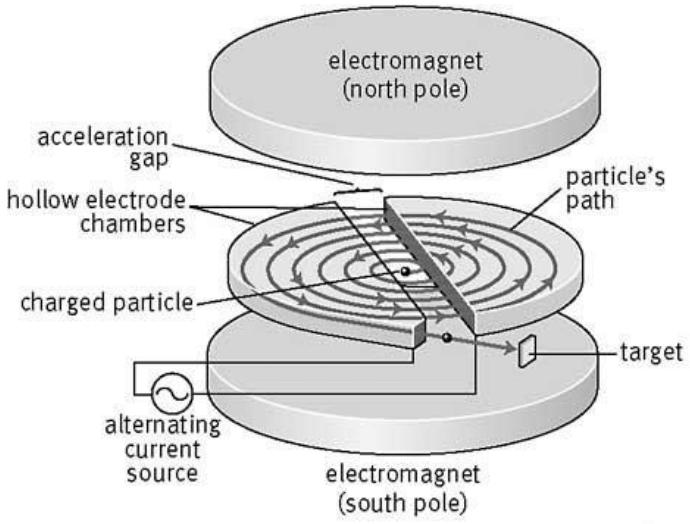


Fig.2.1. Schematic diagram of cyclotron.

Cyclotron and linac are both particle accelerators, but they operate on different principles. Cyclotrons reuse the same acceleration gap multiple times, making them more compact than linacs. Due to their smaller size and simpler design, cyclotrons tend to be less expensive to build and operate compared to linacs. However, the choice between a cyclotron and a linac depends on the specific requirements of the application. Cyclotrons are generally preferred for lower-energy applications, such as isotope production and some types of medical therapy, where compactness, cost-efficiency, and high beam current are essential. Linacs, on the other hand, are better suited for high-energy physics research, cancer therapy requiring high-energy particles, and applications demanding precise beam control.

The cyclotron at TRIUMF is the largest of its kind in the world, measuring almost 18 m (59 feet) in diameter. It produces intense beams of protons of up to 520 MeV.

Working Principle of Cyclotron

- A cyclotron accelerates a charged particle beam using a high frequency alternating voltage which is applied between two hollow "D"-shaped sheet metal electrodes known as the "dees" inside a vacuum chamber.
- The dees are placed face to face with a narrow gap between them, creating a cylindrical space within them for particles to move. Particles are injected into the centre of this space.
- Dees are located between the poles of electromagnet which applies a static magnetic field B perpendicular to the electrode plane.

- The magnetic field causes the path of the particle to bend in a circle due to the Lorentz force perpendicular to their direction of motion.
- An alternating voltage of several thousand volts is applied between the dees. The voltage creates an oscillating electric field in the gap between the dees that accelerates the particles.
- The frequency of the voltage is set so that particles make one circuit during a single cycle of the voltage. To achieve this condition, the frequency must be set to the particle's cyclotron frequency.

The energy of the particles depends on the strength of the magnetic field and the diameter of the dees.

The centripetal force required to keep the particles in a curved path is given by the formula: $$F_c = \frac{mv^2}{R}$$

The force is provided by the Lorentz's force F_B on the magnetic field B

$$F_B = qvB$$

Equating these equations, we get

$$v = \frac{qBR}{m}$$

Hence, the output energy of the particles is given by the expression

$$E_k = \frac{1}{2}mv^2 = \frac{q^2B^2R^2}{2m}$$

Orbital period T is

$$T = \frac{2\pi R}{v} = \frac{2\pi R}{\left(\frac{qBR}{m}\right)} = \frac{2\pi m}{qB}$$

Cyclotron Frequency

$$f = \frac{1}{T} = \frac{qB}{2\,\pi m}$$

A remarkable property of cyclotron frequency is its independence from both the particle's velocity and the radius of its orbit. This means that regardless of how fast a particle is moving or how large its circular path is, as long as the magnetic field strength remains constant, the frequency of its revolution around the field lines remains the same.

For relativistic particle, mass increases $m = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}} = \gamma \mathrm{m}_o$

therefore, frequency decreases.

The kinetic energy $E_k\ (n)$ grows with every passage of the electric field in the gap between the dees by **q.V**

$$E_k\,(n) = E_k + n\ q.V$$

where E_k is the initial kinetic energy of particle leaving the source i.e. $E_k = \frac{1}{2}mv_0^2$ ($v_0\ is$ velocity of particle leaving the source), n = number of passages through the electric field, V Voltage between dees

The velocity v of the particle when it passages through n number of times in the electric field between the "dees":

$$E_k\,(n) = E_k + n\,q.V$$

$$\frac{1}{2}mv^2 = \frac{1}{2}mv_0^2 + n\,q.V$$

$$v = \frac{2}{m}\sqrt{\frac{1}{2}mv_0^2 + n\,q.V}$$

2.3. Procedures:

Turn on the computer and run the simulation. Do as directed in observation copy.

2.4. Observations:

2.4.1 Classical or non-relativistic particle:

Write down the range of

Acceleration voltage (V)……

Magnetic Flux Density B (T) ……

Note: As simulations rely on statistical sampling, it's natural to observe minor differences in outcomes for each run.

a. Set the following values: V = 4000 V and B = 0.1 T, and observe the trajectory of the Proton, Deuteron, and Alpha particle, also note down their kinetic energy (KE).

Trajectory for Proton	Trajectory for Deuteron
K.E. for proton =	K.E. for deuteron =

Trajectory for an alpha particle
K.E. for an alpha particle =

a.Calculate $Vmax$/B ratio (For Proton) (Voltage = 2000 V):

$m_p = 1.67 \times 10^{-27}$ kg
$q_p = 1.6 \times 10^{-19}$ C

Sr. No.	Magnetic Flux Density (T)	$Vmax$ (10^6 m/s)	$Vmax$/B
1.	0.2		
2.	0.3		
3.	0.4		

Comment:

1. At constant voltage, how does the ratio $\boldsymbol{V_{max}}$/B vary with magnetic field B? Write your observation.

2. At constant voltage, how does the R_{max} vary with the magnetic field B? Write your observation.

Calculation for the maximum radius:

$$R_{max} = (mV_{max})/qB$$

b. Note down cyclotron frequency for changing B (For Voltage = 2000 V)

Sr. No.	Magnetic Flux Density (T)	Frequency (MHz)		
		Proton	Deuteron	Alpha
1	0.2			
2	0.3			
3	0.4			

Using the data in the table, determine whether the measured frequencies of the proton, deuteron, and alpha particle at different magnetic flux densities satisfy the relation

$$f = 1/T = qB/2\pi m$$

- Proton ($^1H_1$ nucleus):

$\boldsymbol{m_p \approx 1.67 \times 10^{-27}}$ kg

(For any one observation) $\boldsymbol{f_{Proton}}$ =

- **Deuteron ($^2H_2$ nucleus):**

$\boldsymbol{m_d \approx 3.34 \times 10^{-27}}$ kg

(For any one observation) $\boldsymbol{f_{Deuteron}}$ =

- **Alpha particle ($^4He_2$ nucleus):**

$\boldsymbol{m_\alpha \approx 6.64 \times 10^{-27}}$ kg

(For any one observation) $\boldsymbol{f_{alpha}}$ =

c. Optimizing voltage and magnetic Field to achieve maximum Kinetic Energy

Accelerated Voltage (V)	**Magnetic Flux Density (T)**	**K.E. max**	***n* (number of times particle passage between the "dees")**
2000	0.1		
	0.2		
4000	0.1		
	0.2		

Write your observation.

2.4.2.1 Relativistic without frequency adjustment

For voltage = 100000 V and B (T) = 2 T

Particle	Max velocity attained (10^6 m/s)	Frequency (MHz)
Proton		
Deuteron		
Alpha		

Write your observation.

2.4.2.2 Relativistic with frequency adjustment

For voltage = 100000 V and B (T) = 2 T

Particle	Max Velocity attained (10^6 m/s)	Frequency (MHz)
Proton		
Deuteron		
Alpha		

Write your observation.

2.5 Results:

1. The trajectory of the particles within the cyclotron can be visualized graphically.
2. Study how changes in magnetic field, electric field, and other parameters affect the results.
3. Verify the simulation outcomes with the expected behavior based on cyclotron theory.
4. Adjust parameter to get better cyclotron's performance.
5. Understand the limitations of Cyclotron for smaller charged particle and higher energy particles.

2.5 Precautionary Measure to be taken:

Careful handling of computer and simulation Program.

2.6. Some Sample questions:

1. What role do magnetic and electric fields play in operation of cyclotron?
2. Why is a cyclotron not suitable for accelerating electrons? What limitations does it have in accelerating heavy particles?

3. How does the frequency of the alternating voltage applied to the dees relate to the motion of the particle?
4. Discuss the concept of resonance in a cyclotron. Why is it essential for the efficient acceleration of particles?
5. Explain the relativistic limitations of a cyclotron. How do synchrotrons overcome these limitations?
6. What are the factors that determine the maximum energy attainable by a particle in a cyclotron?
7. How does the design of a cyclotron differ for accelerating protons and heavy ions?
8. Discuss the applications of cyclotrons in medicine, research, and industry.
9. A cyclotron is designed to accelerate protons. If the magnetic field is doubled, how does the cyclotron frequency change?
10. If the radius of the dees in a cyclotron is increased, how does it affect the maximum energy of the accelerated particles?
11. A cyclotron is used to accelerate alpha particles. How would you modify the cyclotron to accelerate protons without changing the magnetic field?
12. A charged particle is injected into a cyclotron at an angle to the magnetic field. Describe the path of the particle.
13. A cyclotron is operating at a fixed frequency. How can the energy of the accelerated particles be increased without changing the frequency?